



Project Title - **Seasonal Agriculture Performance Analysis**

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PROBLEM STATEMENT

- Agricultural activities are influenced by seasonal variations in environmental conditions, farming practices, resource availability and market conditions. As a result, agricultural performance may differ from one season to another. However, raw agricultural data does not clearly explain how agricultural performance changes across seasons or what patterns can be observed in different seasonal conditions. The problem is to analyze the given agricultural dataset and investigate seasonal differences in agricultural performance by identifying meaningful patterns, trends, relationships and variations within the available data.



Project Description

1. Seasonal Agriculture Performance Analysis is a Python-based data analytics project that analyzes agricultural performance across Kharif, Rabi, and Zaid seasons.
2. The project uses 4,000 records and 28 variables covering crops, rainfall, temperature, soil, irrigation, fertilizers, pesticides, yield, production, cost, revenue, profit, and disease/pest risk.
3. Data is cleaned and analyzed using EDA and statistical techniques to identify seasonal patterns, trends, and relationships between agricultural factors.
4. Different visualizations such as bar charts, box plots, distribution plots, and correlation heatmaps are used to present the findings clearly.
5. The analysis found that Kharif had the highest average yield and profit, while Zaid showed negative average profit; drip irrigation also showed higher average yield.
6. Based on the findings, the project provides data-driven recommendations for better crop selection, irrigation management, resource utilization, and seasonal agricultural planning.

WHO ARE THE END USERS?

- **1. Farmers**

- Understand seasonal crop performance
- Compare yield and resource usage
- Support better crop planning

- **2. Agricultural Planners**

- Analyze seasonal agricultural trends
- Support evidence-based planning

- **3. Agricultural Researchers**

- Study relationships between environmental conditions and crop performance
- Identify seasonal patterns

- **4. Government / Agriculture Departments**

- Support agricultural planning and resource management
- Identify areas requiring attention

- **5. Agribusiness & Farm Managers**

- Analyze production, costs and profitability
- Improve resource utilization

Technology Used

- Python
- Pandas
- NumPy
- Matplotlib
- Seaborn
- Jupyter Notebook



RESULTS

[89]: ##Final Seasonal Summary Table

```
final_season_summary = df_clean.groupby("Season").agg({
    "Yield_Tonnes_Ha": "mean",
    "Production_Tonnes": "sum",
    "Rainfall_mm": "mean",
    "Avg_Temperature_C": "mean",
    "Water_Used_m3": "mean",
    "Water_Efficiency_t_per_1000m3": "mean",
    "Total_Cost_INR": "mean",
    "Revenue_INR": "mean",
    "Profit_INR": "mean",
    "Disease_Pest_Risk_pct": "mean"
}).round(2)

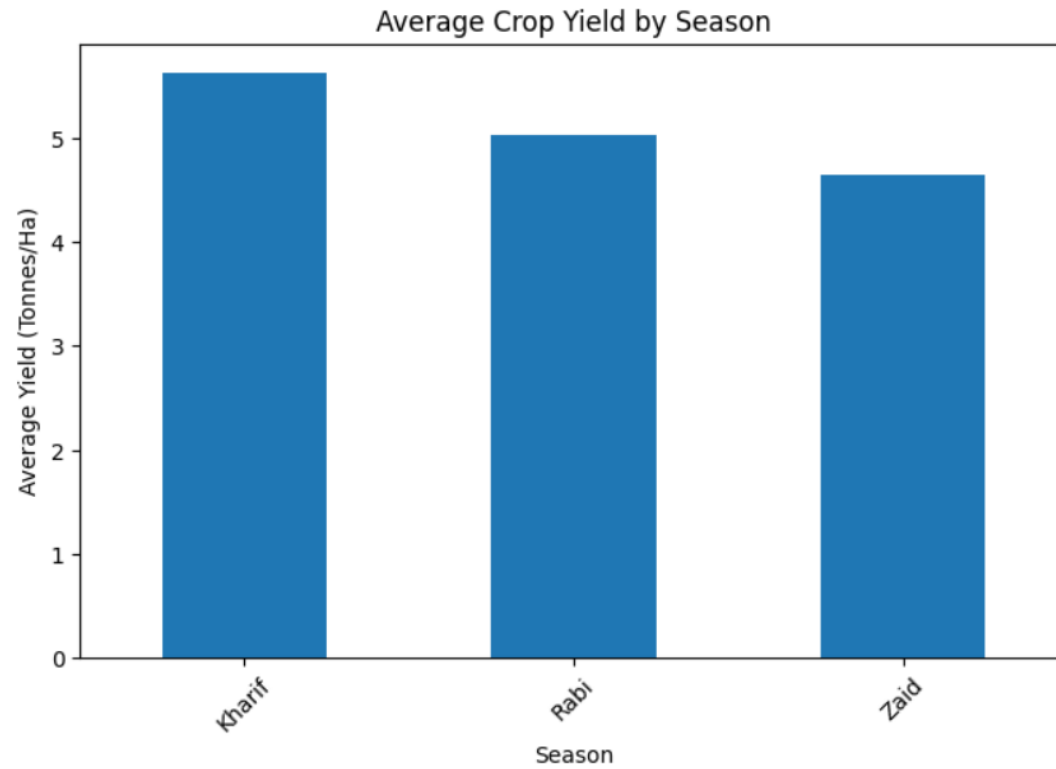
final_season_summary
```

[89]:

	Yield_Tonnes_Ha	Production_Tonnes	Rainfall_mm	Avg_Temperature_C	Water_Used_m3	Water_Efficiency_t_per_1000m3	Total_Cost_INR	Revenue_INR	Profit_INR
Season									
Kharif	5.63	82387.61	849.20	28.45	6102.20	5.89	531804.41	710719.06	178914.6
Rabi	5.04	67498.88	437.62	23.49	5846.99	5.19	513836.58	601526.05	87689.4
Zaid	4.64	23098.35	304.65	31.04	6419.89	4.41	543976.73	519171.90	-24804.8

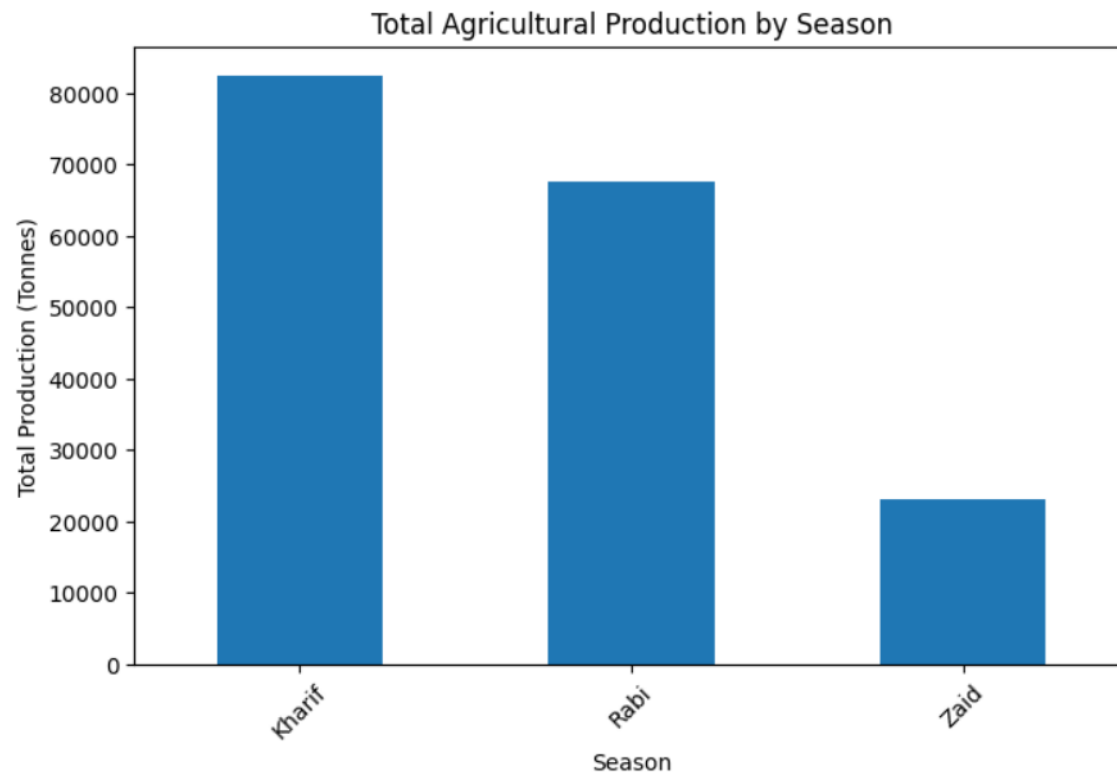
RESULTS

```
[47]: plt.figure(figsize=(8,5))  
  
season_yield.plot(kind="bar")  
  
plt.title("Average Crop Yield by Season")  
plt.xlabel("Season")  
plt.ylabel("Average Yield (Tonnes/Ha)")  
plt.xticks(rotation=45)  
plt.show()
```



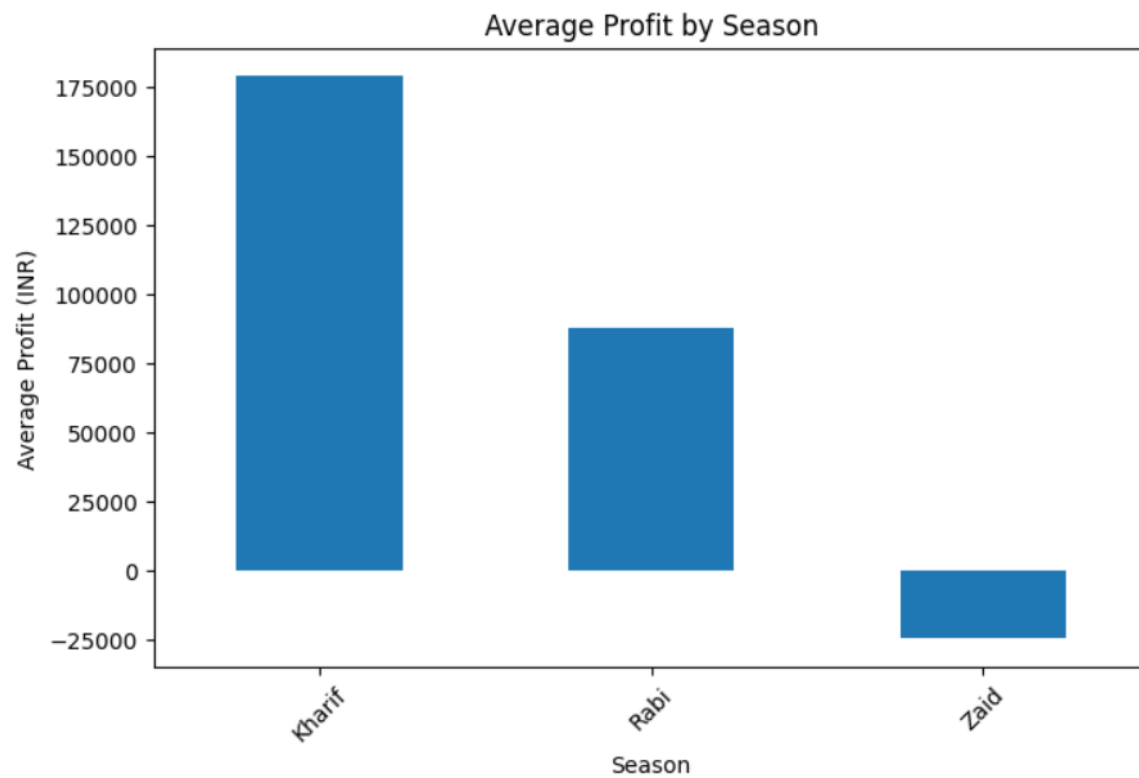
RESULTS

```
[49]: plt.figure(figsize=(8,5))  
  
season_production.plot(kind="bar")  
  
plt.title("Total Agricultural Production by Season")  
plt.xlabel("Season")  
plt.ylabel("Total Production (Tonnes)")  
plt.xticks(rotation=45)  
plt.show()
```



RESULTS

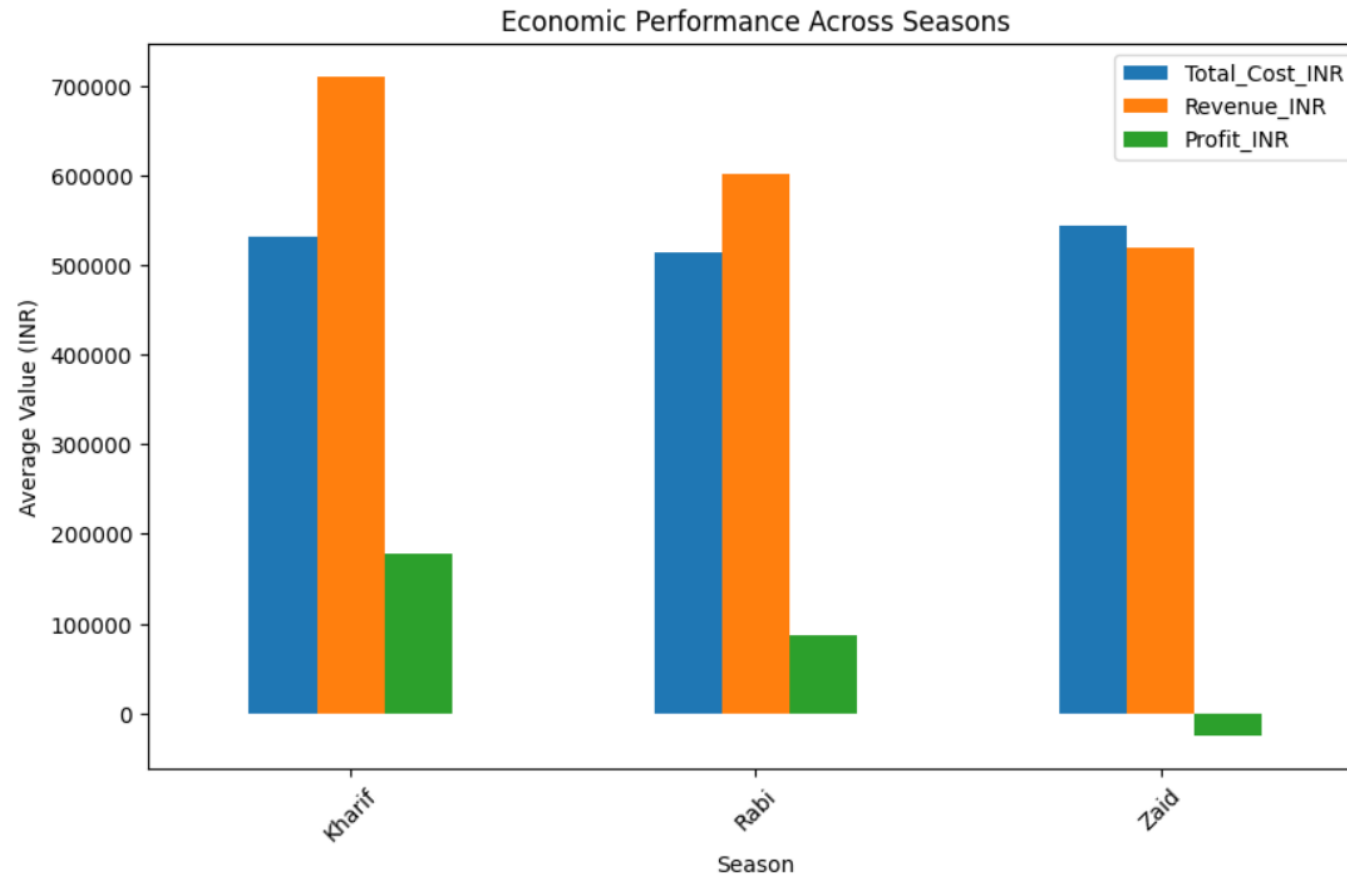
```
[51]: plt.figure(figsize=(8,5))  
  
      season_profit.plot(kind="bar")  
  
      plt.title("Average Profit by Season")  
      plt.xlabel("Season")  
      plt.ylabel("Average Profit (INR)")  
      plt.xticks(rotation=45)  
      plt.show()
```



RESULTS

```
[53]: economic_summary.plot(kind="bar", figsize=(10,6))

plt.title("Economic Performance Across Seasons")
plt.xlabel("Season")
plt.ylabel("Average Value (INR)")
plt.xticks(rotation=45)
plt.legend()
plt.show()
```

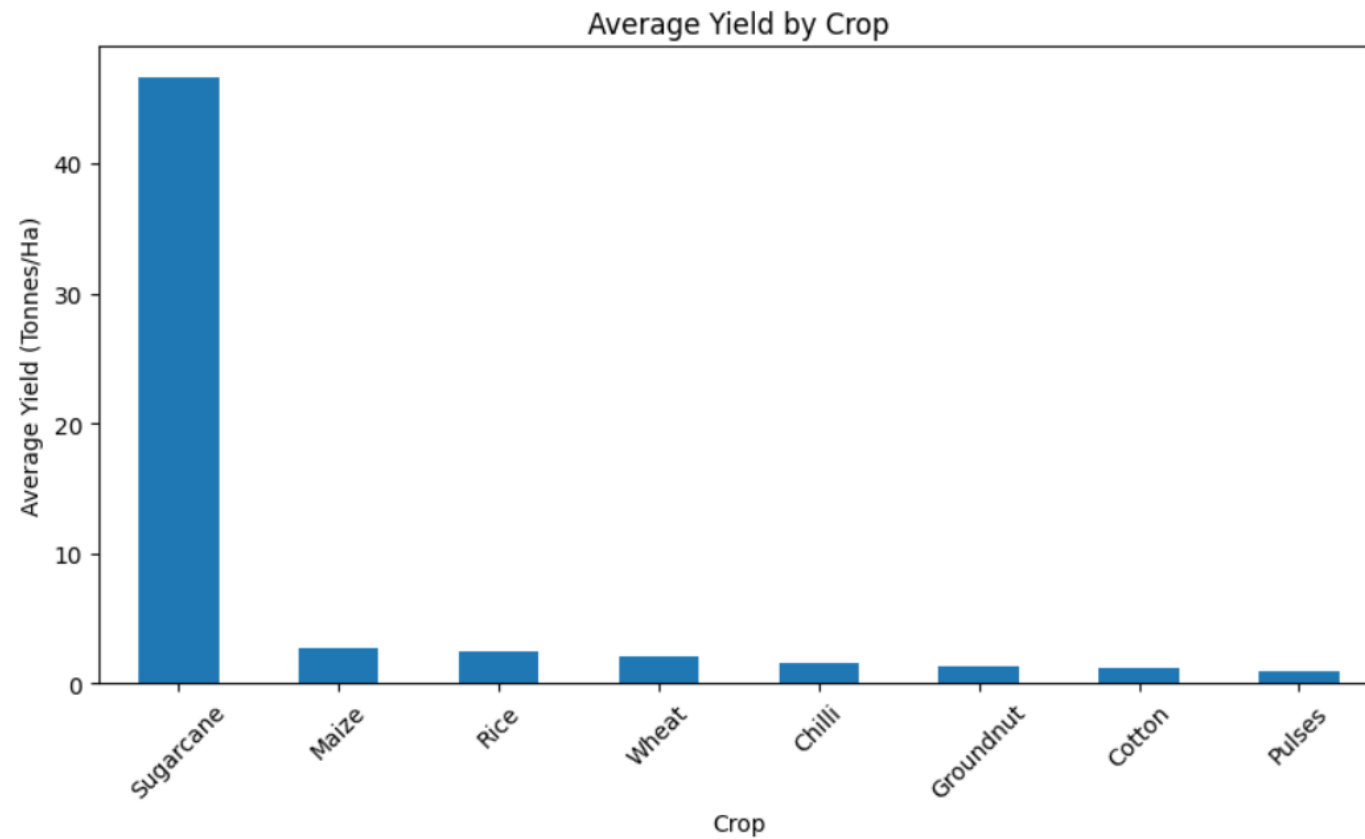


RESULTS

```
[55]: plt.figure(figsize=(10,5))

crop_yield.plot(kind="bar")

plt.title("Average Yield by Crop")
plt.xlabel("Crop")
plt.ylabel("Average Yield (Tonnes/Ha)")
plt.xticks(rotation=45)
plt.show()
```



RESULTS

```
[57]: plt.figure(figsize=(10,6))

sns.heatmap(crop_season_yield, annot=True, fmt=".2f")

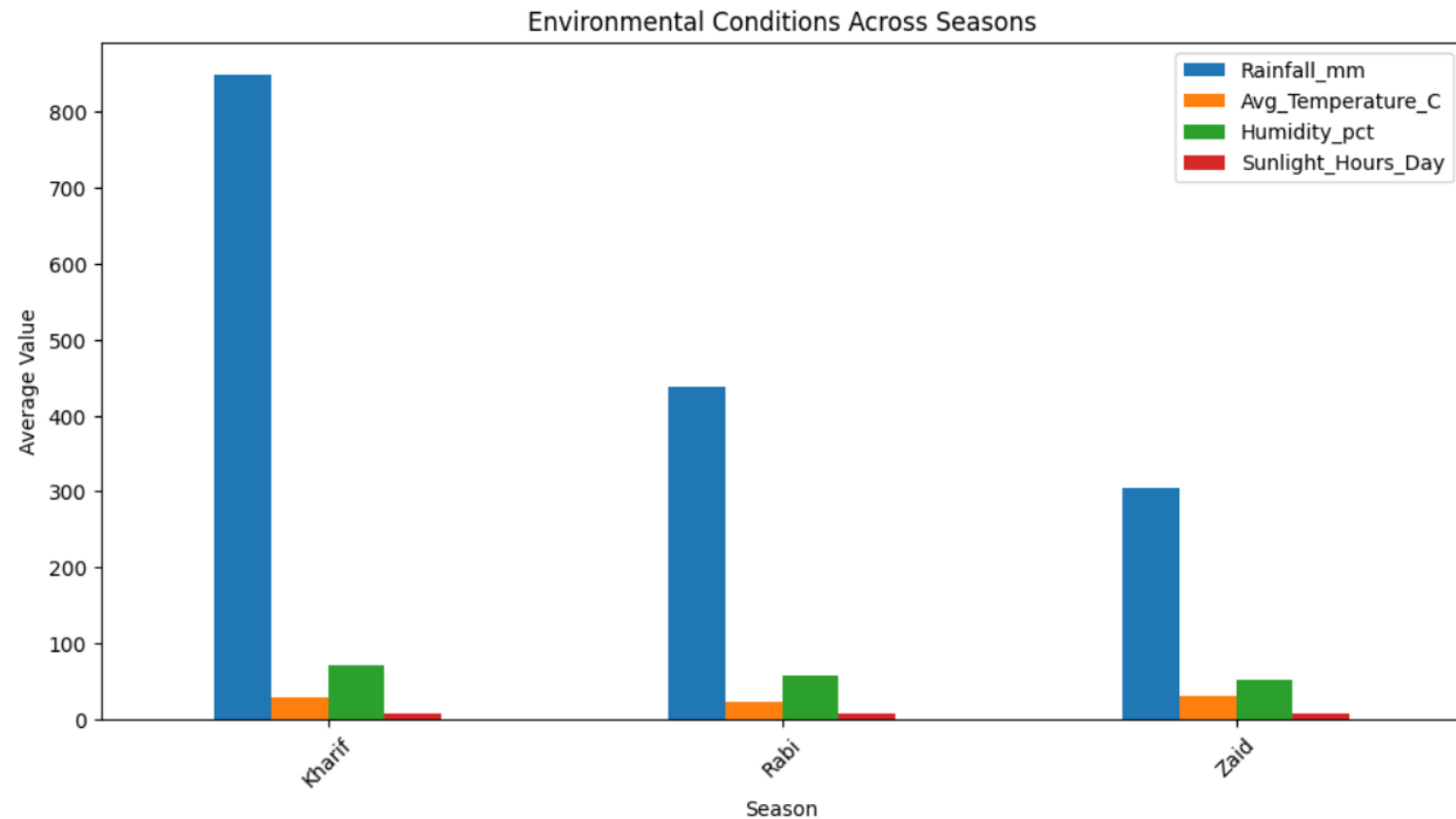
plt.title("Average Crop Yield Across Seasons")
plt.xlabel("Season")
plt.ylabel("Crop")
plt.show()
```



RESULTS

```
[59]: environment_summary.plot(kind="bar", figsize=(12,6))
```

```
plt.title("Environmental Conditions Across Seasons")  
plt.xlabel("Season")  
plt.ylabel("Average Value")  
plt.xticks(rotation=45)  
plt.legend()  
plt.show()
```

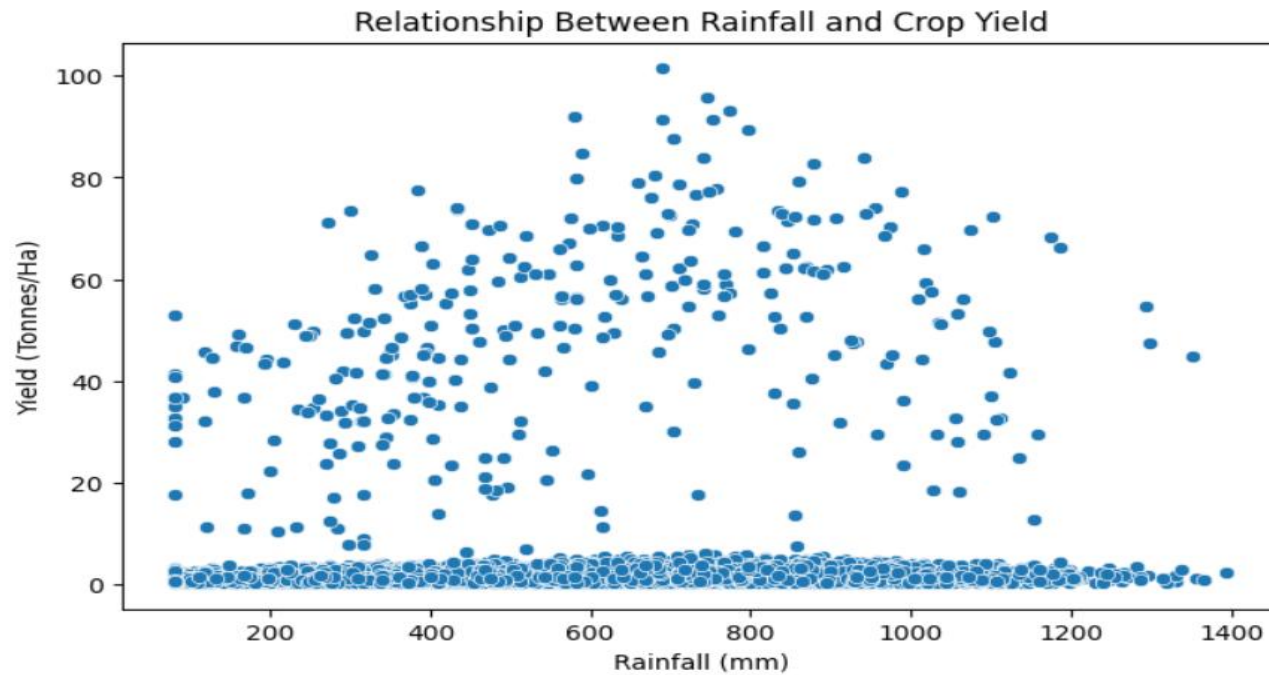


RESULTS

```
[60]: plt.figure(figsize=(8,5))

sns.scatterplot(
    data=df_clean,
    x="Rainfall_mm",
    y="Yield_Tonnes_Ha"
)

plt.title("Relationship Between Rainfall and Crop Yield")
plt.xlabel("Rainfall (mm)")
plt.ylabel("Yield (Tonnes/Ha)")
plt.show()
```

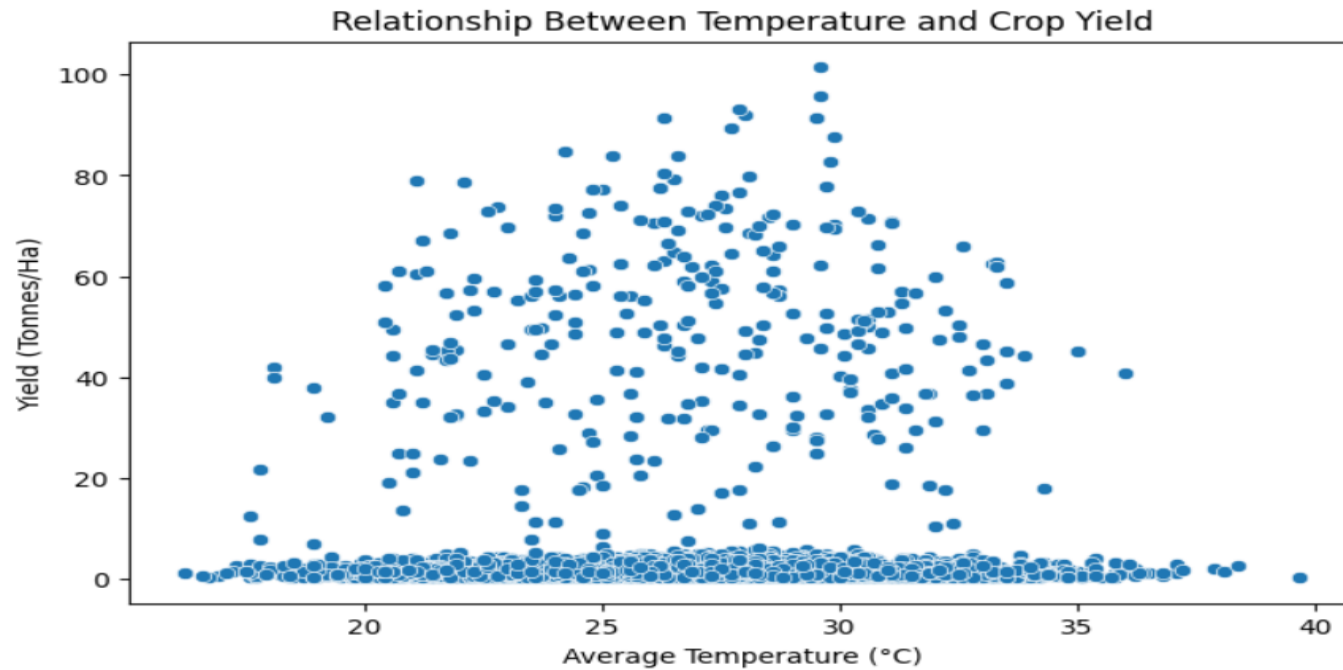


RESULTS

```
[61]: plt.figure(figsize=(8,5))

sns.scatterplot(
    data=df_clean,
    x="Avg_Temperature_C",
    y="Yield_Tonnes_Ha"
)

plt.title("Relationship Between Temperature and Crop Yield")
plt.xlabel("Average Temperature (°C)")
plt.ylabel("Yield (Tonnes/Ha)")
plt.show()
```

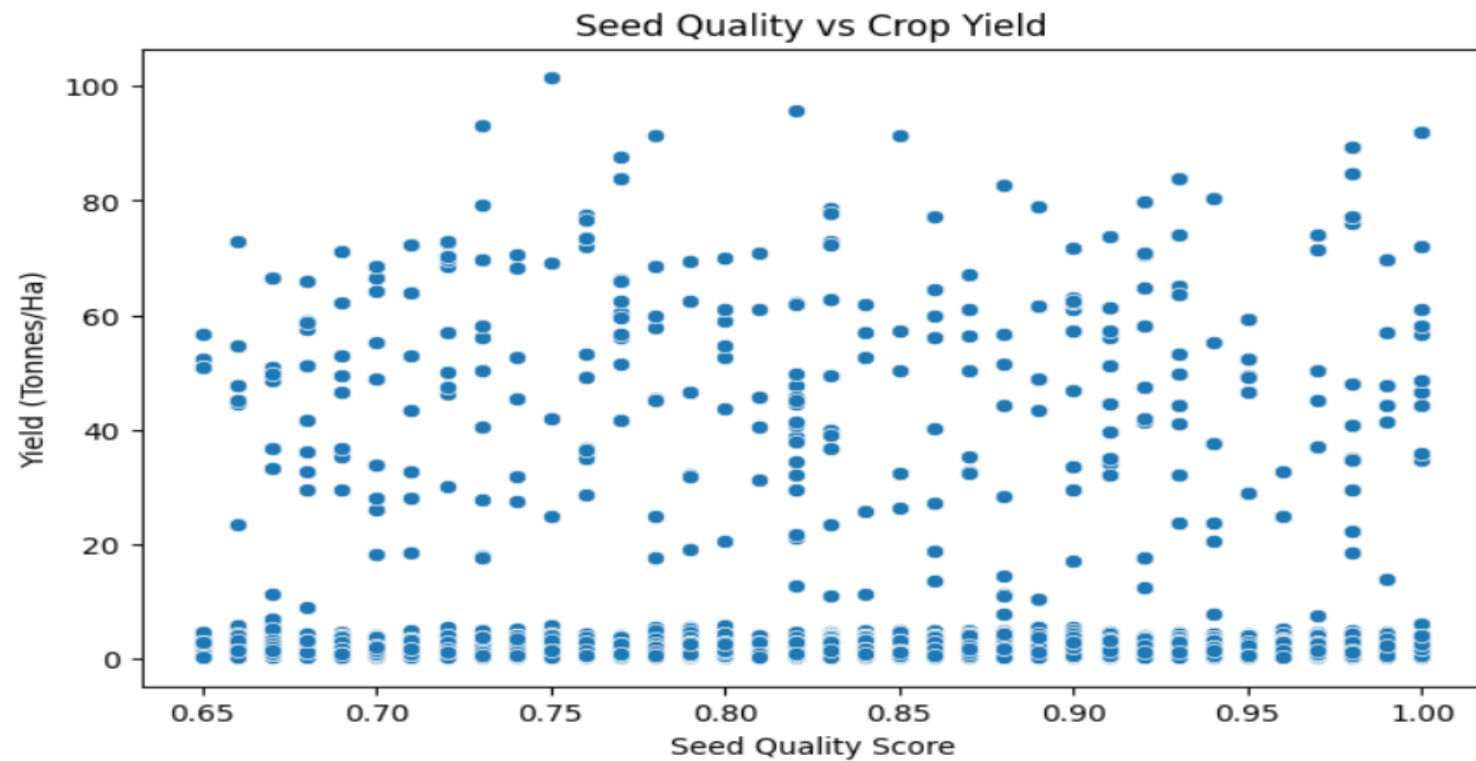


RESULTS

```
] plt.figure(figsize=(8,5))

sns.scatterplot(
    data=df_clean,
    x="Seed_Quality_Score",
    y="Yield_Tonnes_Ha"
)

plt.title("Seed Quality vs Crop Yield")
plt.xlabel("Seed Quality Score")
plt.ylabel("Yield (Tonnes/Ha)")
plt.show()
```

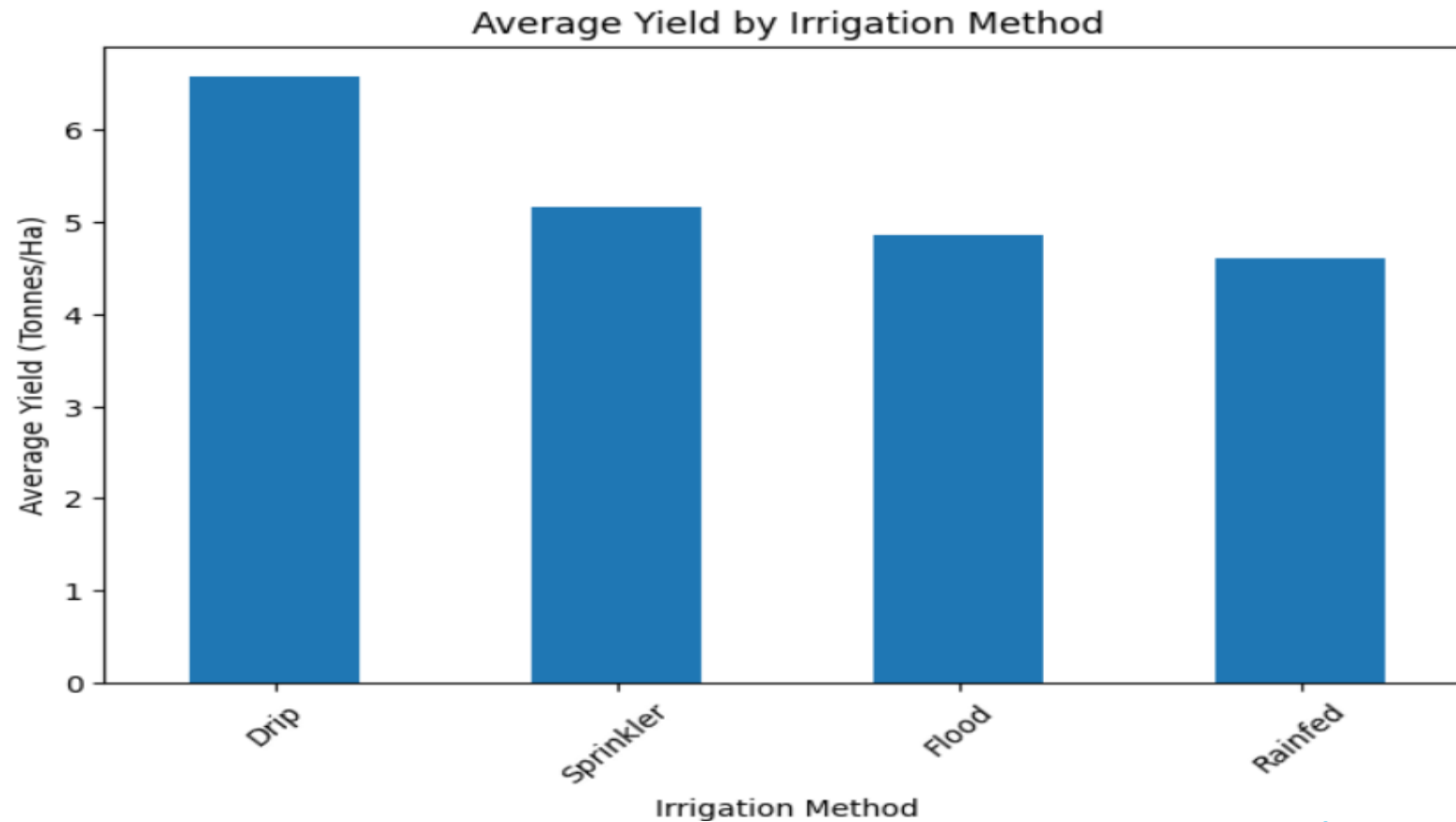


RESULTS

```
[6]: plt.figure(figsize=(8,5))

irrigation_yield.plot(kind="bar")

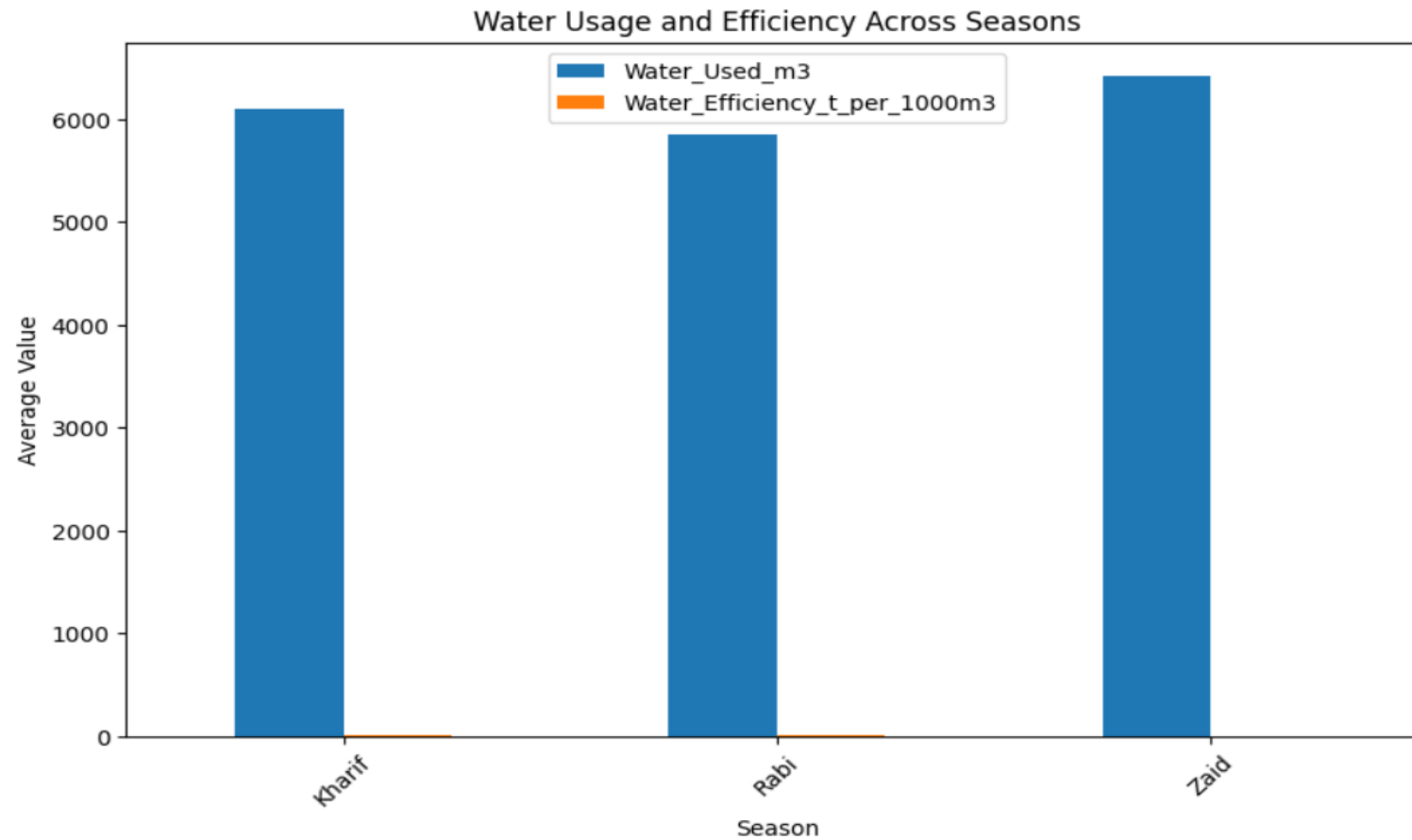
plt.title("Average Yield by Irrigation Method")
plt.xlabel("Irrigation Method")
plt.ylabel("Average Yield (Tonnes/Ha)")
plt.xticks(rotation=45)
plt.show()
```



RESULTS

```
|: water_summary.plot(kind="bar", figsize=(10,6))

plt.title("Water Usage and Efficiency Across Seasons")
plt.xlabel("Season")
plt.ylabel("Average Value")
plt.xticks(rotation=45)
plt.show()
```

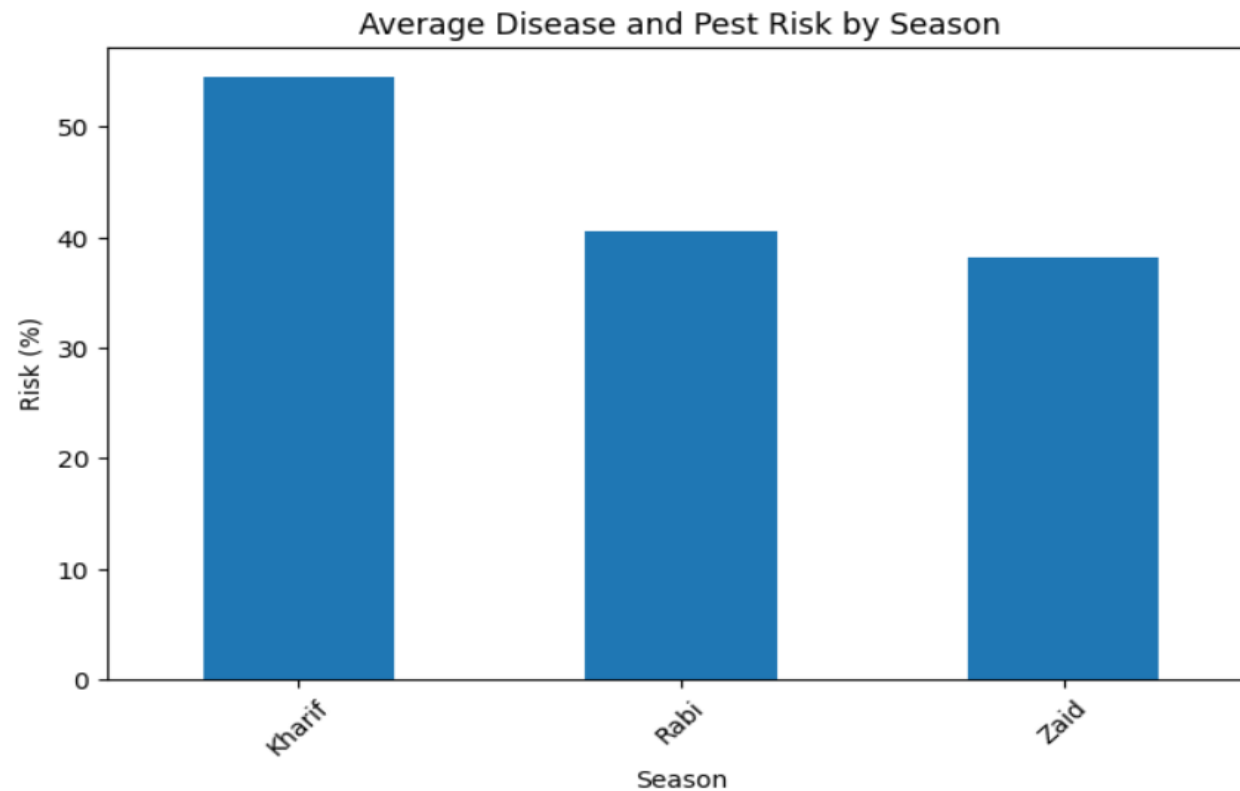


RESULTS

```
[73]: plt.figure(figsize=(8,5))

risk_season.plot(kind="bar")

plt.title("Average Disease and Pest Risk by Season")
plt.xlabel("Season")
plt.ylabel("Risk (%)")
plt.xticks(rotation=45)
plt.show()
```



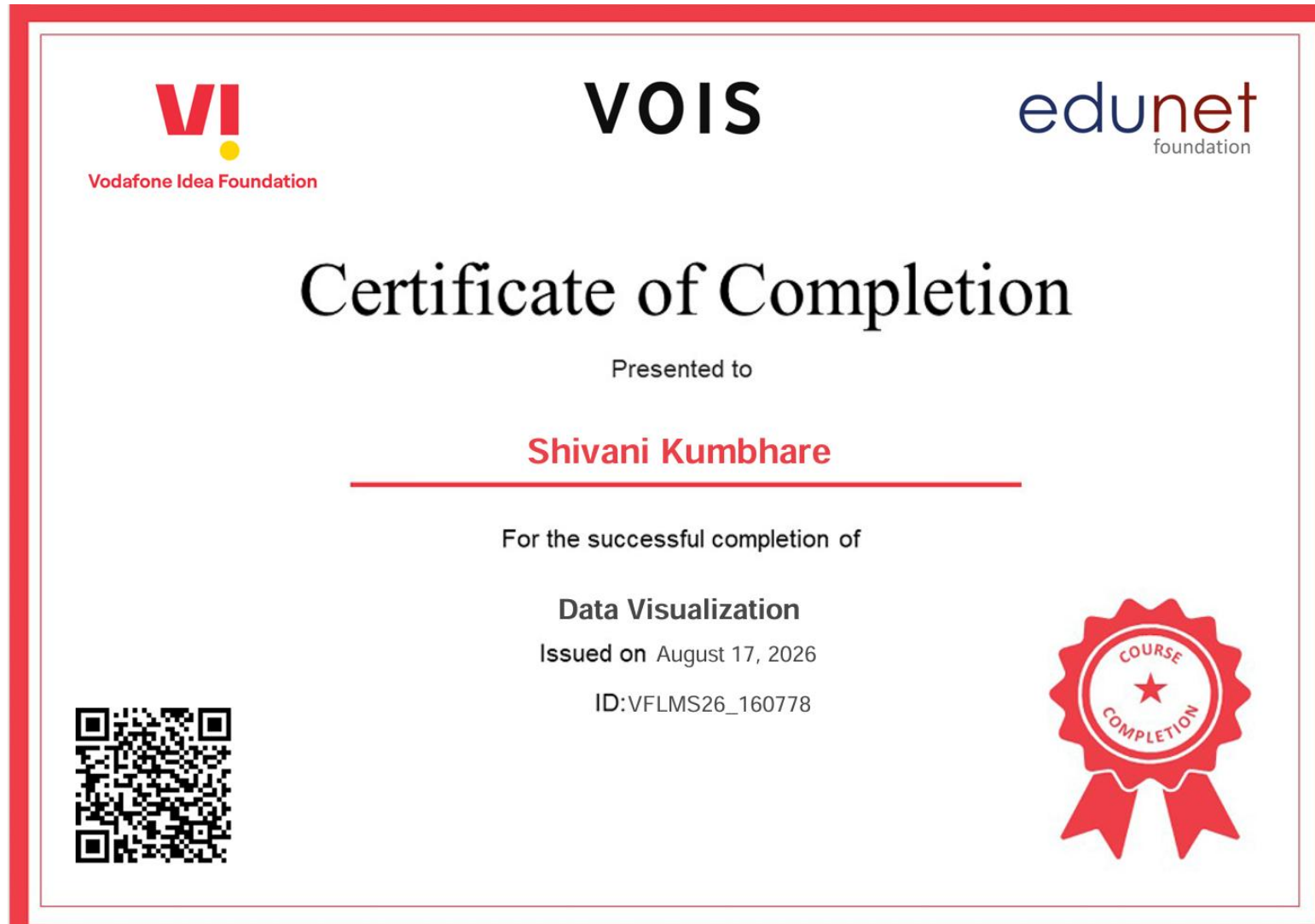
Future scope

- Machine Learning-based yield prediction
- Crop recommendation system
- Weather data integration
- Seasonal forecasting
- Interactive dashboard
- Real-time agricultural monitoring

GitHub Link

<https://github.com/shivanikumbhare-01/Seasonal-Agriculture-Performance-Analysis/tree/main>

VOIS Course completion certificate - Course name- Data Visualization



Thank you